

Reduction and Echelon Forms

Reduction

We solve a linear system by performing a series of row operations on the augmented matrix until the solution is easy to see.

It is often called ***Gaussian Elimination*** or ***Gauss-Jordan Elimination***.

Echelon Form

Defn. An unaugmented matrix is in **echelon form** if:

- (a) for each row that is not completely zero, the first nonzero entry has zeros below it and below its preceding zeroes; and
- (b) every all-zero row is below every other row.

Example of Echelon Form

An example echelon form is

$$\begin{bmatrix} -7 & 5 & 0 & -10 \\ 0 & 0 & 1 & \frac{1}{5} \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

An augmented system in echelon form can be solved by back-substitution, working upwards in the matrix.

Row Reduction to Echelon Form

A **pivot** is the first nonzero entry in a row.

ALGOR *Achieving Echelon Form*

- Go to first nonzero column (but not augmented column). If needed, interchange rows so top entry in column is not zero.
- Add suitable multiples of first row to create zeroes below pivot.
- Repeat, ignoring rows with pivots, until no nonzero row.

Example of Achieving Echelon Form

Consider matrix

$$\left[\begin{array}{ccc|c} -1 & 2 & -1 & 0 \\ 0 & 2 & -8 & 8 \\ 6 & -2 & -4 & 10 \end{array} \right]$$

The first nonzero column is the first column. The top entry is nonzero; so no interchange needed. The -1 is the pivot.

The second entry in the column is 0; so no work needed. The third entry is 6. One adds 6

times the first row to the third row:

$$\left[\begin{array}{ccc|c} \textcircled{-1} & 2 & -1 & 0 \\ 0 & 2 & -8 & 8 \\ 6 & -2 & -4 & 10 \end{array} \right] R'_3 = R_3 + 6R_1 \rightsquigarrow \left[\begin{array}{ccc|c} -1 & 2 & -1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & 10 & -10 & 10 \end{array} \right]$$

Next the second column (ignoring first row). There is nonzero where needed: so 2 is the pivot. To get zero in third row, subtract 5 times the second row from the third row. And echelon form is reached.

$$\left[\begin{array}{ccc|c} -1 & 2 & -1 & 0 \\ 0 & \textcircled{2} & -8 & 8 \\ 0 & 10 & -10 & 10 \end{array} \right] R'_3 = R_3 - 5R_2 \rightsquigarrow \left[\begin{array}{ccc|c} -1 & 2 & -1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & 0 & 30 & -30 \end{array} \right]$$

driver: 8

Reduced Row Echelon Form

Defn. An unaugmented matrix in echelon form is in **reduced row echelon form** if furthermore

(c) the pivot in every row is a 1 and has zeroes above it.

Fact. Every matrix is row-equivalent to a unique matrix in reduced row echelon form.

Achieving Reduced Row Echelon Form

ALGOR ***Achieving Reduced Row Echelon Form***

From right to left, add suitable multiples of the rows to create zeroes above each pivot and also make pivots 1 by scaling.

Example

We earlier reached an echelon form of

$$\left[\begin{array}{ccc|c} -1 & 2 & -1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & 0 & 30 & -30 \end{array} \right]$$

The reduction continues with scaling the third row by dividing by 30, followed by adding 8 times it to the second row, and then adding 1 times it to the first row to get

$$\left[\begin{array}{ccc|c} -1 & 2 & -1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & 0 & \textcircled{30} & -30 \end{array} \right] \begin{array}{l} R'_1 = R_1 + R'_3 \\ R'_2 = R_2 + 8R'_3 \\ R'_3 = R_3/30 \end{array} \rightsquigarrow \left[\begin{array}{ccc|c} -1 & 2 & 0 & -1 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{array} \right]$$

There after we do the same for the second column

$$\left[\begin{array}{ccc|c} -1 & 2 & 0 & -1 \\ 0 & \textcircled{2} & 0 & 0 \\ 0 & 0 & 1 & -1 \end{array} \right] \begin{array}{l} R'_1 = R_1 - 2R'_2 \\ R'_2 = R_2/2 \end{array} \rightsquigarrow \left[\begin{array}{ccc|c} -1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{array} \right]$$

and finally, only scaling is needed in the first

row to produce reduced row echelon form.

$$\left[\begin{array}{ccc|c} \textcircled{-1} & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{array} \right] R'_1 = -R_1 \quad \rightsquigarrow \quad \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{array} \right]$$

This system has unique solution $x_1 = 1$, $x_2 = 0$,
and $x_3 = -1$.

Recognizing Consistent System

Fact. *A linear system is consistent if and only if the echelon form of the augmented matrix has no row that is all zeroes except for the augmented column.*

Solutions of Linear Systems

Defn. A **basic** variable is one whose column has a pivot; otherwise it is a **free** variable.

ALGOR The **parametric description** of the solution set is obtained by expressing each basic variable in terms of the free variables.

Example of Parametric Description

Consider the row reduced echelon form:

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & -3 & -1 \\ 0 & 0 & 1 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

The system is consistent. Assuming variables $x_1, \dots, x_4$, the free variables are x_2 and x_4 . So system has solution

$$\begin{aligned} x_1 &= -1 - 2x_2 + 3x_4 \\ x_3 &= 5 - x_4 \end{aligned}$$

Summary

In echelon form, the pivot (first nonzero entry) in each row has zeros below it and its preceding zeroes; in reduced row echelon form, the pivot is a 1 and also has zeroes above it.

For echelon form, for each column get pivot in needed row and then add suitable multiples of row to create zeroes below pivot. For reduced row echelon form, for each column add suitable multiples of that row to create zeroes above the pivot and also make pivot 1.

Summary (cont)

The system is inconsistent if the echelon form has a row that is all zeroes except for the augmented column.

A basic variable is one whose column has a pivot; otherwise it is a free variable. The parametric description of the solution set is obtained by expressing each basic variable in terms of the free variables.